

Scalability & Future Plan

Date	15 March 2026
Team ID	
Project Name	Smart Lender :AI/ ML-Based Loan Approval Prediction System

Scalability & Future Plan:

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited to a single trained dataset and predefined input features.	Prediction accuracy may decrease when processing diverse real-world applicant profiles.	High
2	Basic Flask application with limited concurrent user support.	Performance may degrade when multiple users access the application simultaneously.	Medium
3	No user authentication, database integration, or loan application history.	Limits security, scalability, and long-term record management.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of concurrent users.	Deploy on scalable cloud infrastructure (IBM Cloud/Kubernetes) with load balancing and auto-scaling to handle large numbers of users.
2	Data Storage	Uses local files and a serialized machine learning model without persistent storage.	Integrate a cloud-based relational database (such as MySQL or PostgreSQL) to securely store applicant records, prediction history, and user information.
3	Performance	Single-instance Flask application with one prediction model.	Optimize model inference, enable caching, and use production servers (Gunicorn/Nginx) to improve response time and throughput.
4	Security	Basic application security with limited access control.	Implement user authentication, role-based access control, HTTPS encryption, secure

			API management, and regular security audits to protect sensitive financial data.
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Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Implement user authentication, role-based access, and database integration to securely manage loan applications and user records.	3–6 Months	Improves security, data management, and supports multiple users.
Phase 3	Integrate advanced AI features such as explainable AI (XAI), credit score analysis, and document verification using OCR.	6–12 Months	Increases prediction transparency, improves decision-making, and reduces manual verification effort.
Phase 4	Deploy a scalable cloud-based solution with mobile application support, API integration with banking systems, and real-time analytics dashboards.	12–18 Months	Enhances scalability, accessibility, and enables enterprise-level adoption by financial institutions.